

CURRENT IMPLEMENTATION REPORT

LearnerOS

Product experience, assessment context,
and insight-generation boundaries

REFERENCE	SCOPE	STATUS
10 August 2026	Live learner UI + relevant assessment paths	Code-grounded; unrelated tooling excluded

What the current product does

LearnerOS presents an NCERT-aligned learning journey from grade and subject selection through chapter reading, interactive concept experiences, assessment, personalized insights, and tutoring. The current evidence shows a coherent learner-facing product rather than a collection of disconnected prototypes.

Learn

Navigate grade, subject, chapter, section, and subsection content. Use concept labs and chapter knowledge graphs to move between explanation and exploration.

Assess

Generate written and multiple-choice questions for the active subsection. The entire parent section supplies supporting context, but the target subsection remains the assessment boundary.

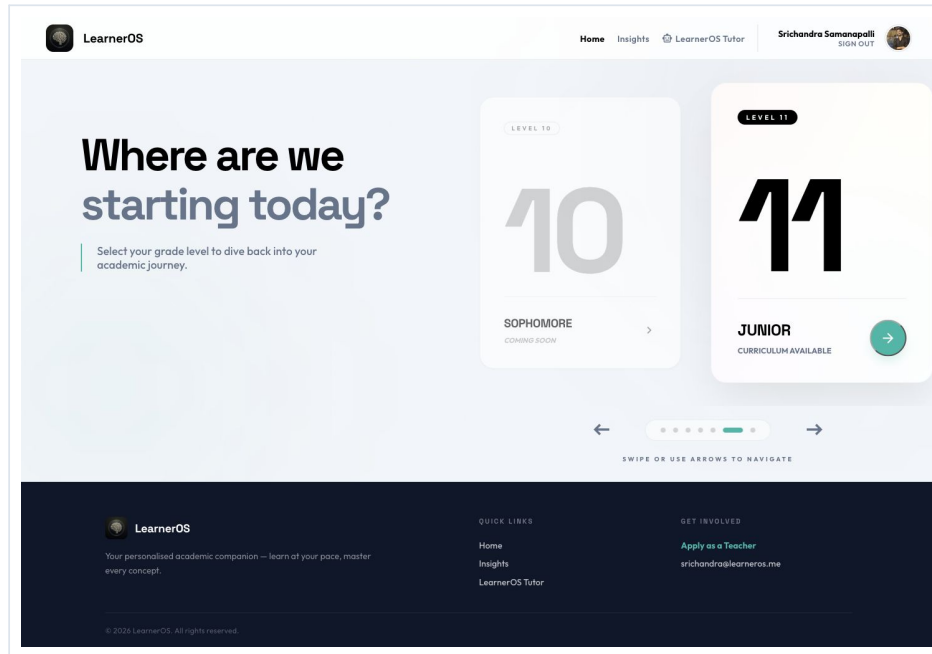
Adapt

Evaluate the learner response, then use curriculum concepts as labels and evidence anchors for insights. If the section has no direct concepts, chapter concepts can supply the fallback scope.

The assessment boundary

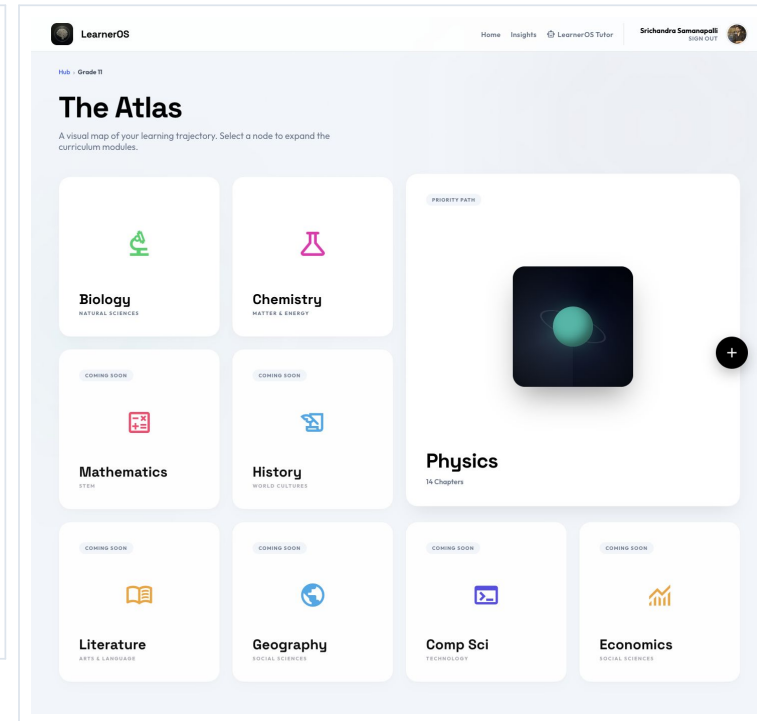
STAGE	INPUT THAT SHOULD CONTROL IT	CONCEPT ROLE
Question generation	The requested subsection is primary. The full parent section is reference context when the subsection alone is too narrow.	No concept link is required to create the question. Linked concepts must not broaden the tested material beyond the target subsection.
Answer evaluation	Question, learner answer, target subsection content, and the parent section for interpretation.	Use direct section concepts when present; otherwise use chapter concepts as the insight/evidence scope.
Insight creation	Only evidence supported by the learner response and assessment context.	Concepts categorize and connect the evidence. They do not retroactively redefine what the question tested.

From entry point to curriculum



Grade entry

The home experience presents available grade levels and routes the learner into the curriculum. Grade 11 is shown as available while Grade 10 remains clearly marked as coming soon.



Subject atlas

The grade view shows subject availability and identifies Physics as the active priority path with 14 chapters. Other subjects remain visible without implying they are fully released.

Chapter selection and section reading

Physics 14 Chapters

- CHAPTER 01 Units and Measurement 12 Sections · 17 Exercises
- CHAPTER 02 Motion in a Straight Line 7 Sections · 18 Exercises
- CHAPTER 03 Motion in a Plane 12 Sections · 22 Exercises
- CHAPTER 04 Laws of Motion 14 Sections · 23 Exercises

Thermal Properties of Matter

Chapter 10 - Physics

CHAPTER OVERVIEW	SECTIONS	12
This chapter delves into the thermal properties of matter, starting with the fundamental definitions of temperature and heat, and explaining how temperature is measured using different scales (Celsius, Fahrenheit, Kelvin) and gas thermometers. The concept of	EXERCISES	20
	CONCEPTS	5

Overview of Thermal Properties

Subsection 1 of 1

We have common sense notions of heat and temperature. Temperature is a measure of 'hotness' of a body; for instance, a kettle with boiling water is hotter than a box containing ice. In physics, we define these notions more carefully. This chapter explores what heat is, how it is measured, and the various processes by which heat flows from one body to another. It addresses practical phenomena such as why blacksmiths heat iron rings before fitting them on wooden wheels, why sea breezes reverse direction after sunset, and why the temperature of water remains constant during boiling or freezing despite continuous heat flow.

100% 1 of 1

Knowledge Graph

Chapter context

The Physics view combines chapter navigation, cover imagery, an overview, and counts for sections, exercises, and concepts. This makes curriculum structure visible before the learner enters a lesson.

Immediate learning loop

Within a section, the learner can read the active subsection and move directly to concepts, a written test, MCQ practice, chapter exercises, a section tutor, or the knowledge graph.

Two modes, one content boundary

The screenshot shows the 'Testing Your Understanding' interface. At the top, the LearnerOS logo and navigation links (Home, Insights, LearnerOS Tutor) are visible, along with the user's name 'Srichandra Samanapalli' and a 'SIGN OUT' button. The main heading is 'Testing Your Understanding' with a subtext: 'Take your time to provide a detailed explanation. Our AI will analyze your response for accuracy and depth.' Below this, there's a section for 'AI GENERATED QUESTION' with a paragraph about heat flow and temperature change. To the right, there's a 'Context' section with 'SUBJECT & UNIT' (Physics - Chapter 10) and 'SECTION' (Introduction). Below that, 'Key Terms' are listed: heat, temperature, hotness, heat flow, boiling, freezing, and thermal expansion. A 'Pro Tip' box provides additional context. At the bottom, there's a large text area for 'Your Detailed Answer' with a prompt: 'Start typing your detailed answer here... Explain the concepts thoroughly and connect key ideas.' A 'Cancel' button and a 'Submit for AI Evaluation' button are at the bottom right.

Written assessment

A generated open-response question asks the learner to explain or connect ideas. The page provides section context, key terms, and a detailed-answer field before AI evaluation.

The screenshot shows the 'The Practice' interface for a Multiple Choice Question (MCQ). At the top, the LearnerOS logo and navigation links (Home, Insights, LearnerOS Tutor) are visible, along with the user's name 'Srichandra Samanapalli' and a 'SIGN OUT' button. The main heading is 'The Practice' with a subtext: 'SECTION 10.1 - MCQ'. A 'Back to Lesson' button is at the top right. The question is 'Question 1 of 2' and the progress is '0% Complete'. The question text is: 'A student observes that during the boiling of water, the temperature remains constant at 100°C even though heat is being continuously supplied to the water. Which of the following best explains this phenomenon in the context of thermal properties?'. There are four options: A. The heat being supplied is lost to the surroundings at the same rate, so the net energy input is zero. B. The heat supplied is entirely used to change the phase of water from liquid to vapour, rather than raising its temperature. C. The temperature of water cannot exceed 100°C because water has a fixed maximum temperature limit. D. The thermometer used to measure the temperature is inaccurate and fails to register the rising temperature. At the bottom, there are 'Skip' and 'Submit Answer' buttons.

MCQ practice

The practice view presents a bounded question set with selectable options, progress state, skip behavior, and answer submission. The visible example remains tied to the active thermal-properties material.

How content and concepts are separated

The earlier ambiguity came from treating concept linkage as though it controlled both question generation and insight creation. The current implementation keeps those responsibilities separate.

1. Resolve target

Validate that the requested subsection belongs to the section. The requested subsection ID remains authoritative even if a model echoes another ID.

2. Generate

Use target subsection content as the primary assessment source. Pass the complete parent section only as supporting context for missing local context.

3. Evaluate

Judge the response against the target subsection. Do not require facts that appear only in sibling subsections.

4. Select concept scope

Use directly linked section concepts when they exist. If the section has none, use concepts attached elsewhere in the same chapter.

5. Create insight

Create only evidence supported by the response, then attach it to the applicable concept scope and subsection/source.

6. Preserve boundaries

A chapter-level concept fallback helps classify insight evidence; it must not cause the generated question to test unrelated chapter material.

Example: section 10.3

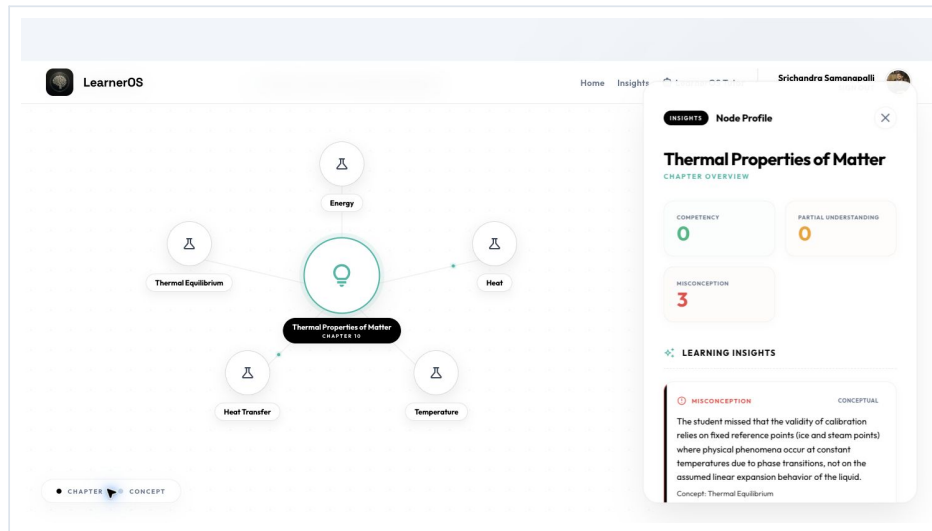
No direct section concepts

Use chapter concepts during answer evaluation and insight creation. Continue generating the question only from the active subsection, with its parent section as context.

Implementation evidence

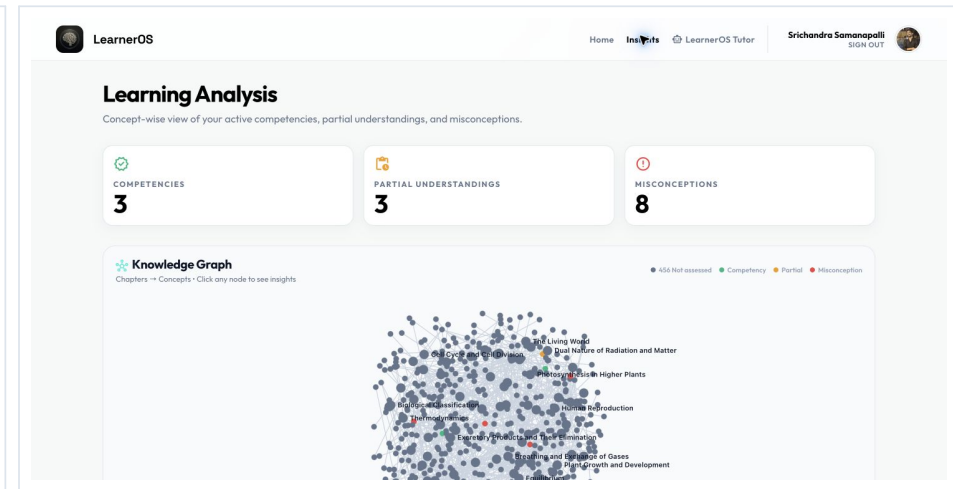
Relevant source paths: `backend/app/routers/test.py` for assessment context, generation, evaluation, and insight persistence; `backend/app/routers/curriculum.py` for section/subsection and concept reads. Evaluation and observability tooling that does not change this product behavior is intentionally outside this report.

Insights are visible at concept and curriculum scale



Node-level evidence

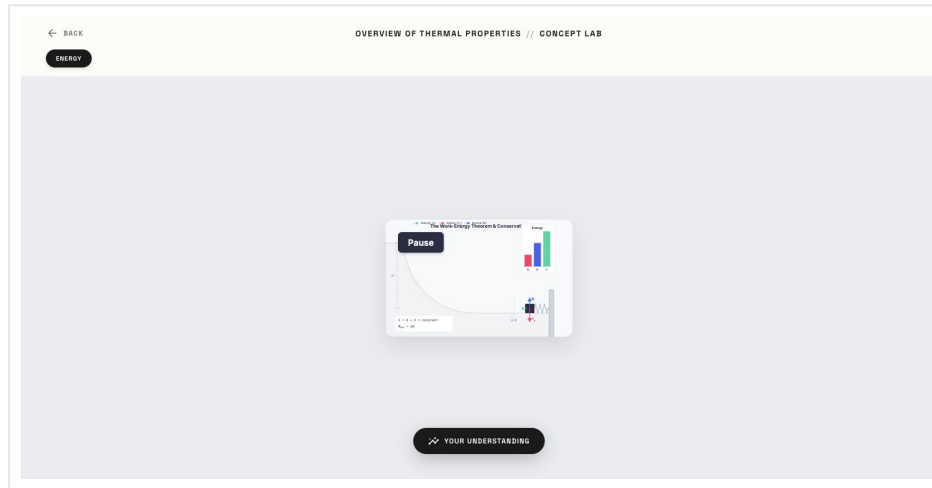
The chapter knowledge graph exposes linked concepts and a node profile that summarizes competencies, partial understanding, and misconceptions with the underlying learning insight.



Learner-wide analysis

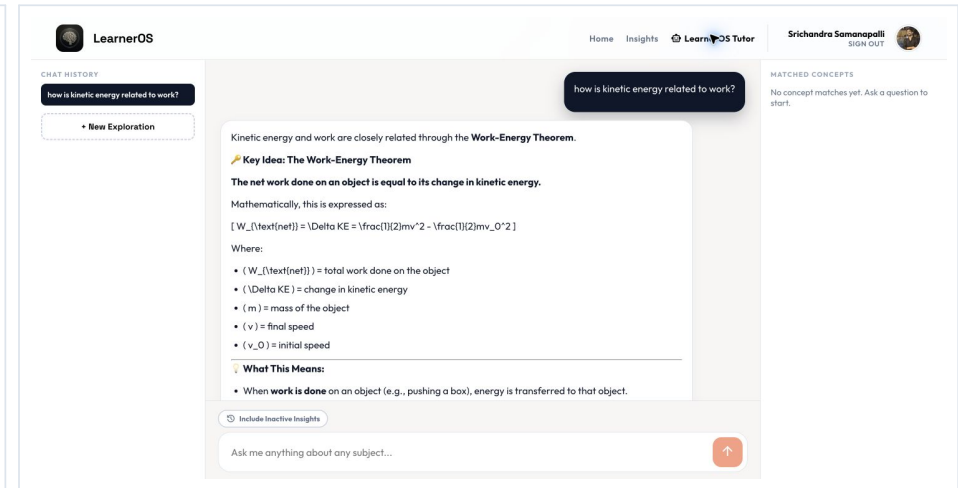
The Insights dashboard aggregates active evidence across the curriculum and visualizes assessed versus not-assessed concepts. This gives the learner a broader view without losing concept-level drill-down.

Interactive concepts and AI tutoring



Concept lab

A dedicated environment can host an interactive visual or simulation associated with a concept. The learner can pause the experience and move into a self-check through the understanding control.



LearnerOS Tutor

The tutor supports open-ended questions with conversation history and concept matching. The visible response explains the Work-Energy Theorem and keeps the learner in an ongoing exploration thread.

Learner control and teacher access

Profile settings

Learners can associate an institute, manage grade preference, and delete their profile. The destructive action is visibly separated in a danger zone.

Verified teacher access

Teacher access is application-based and institute-scoped. The form requests department, verification links, and a statement of purpose before access to analytics and learner-risk views.

What this report confirms - and what it does not

Confirmed from the captured product

The learner can navigate the visible Grade 11 curriculum, read section content, open concept experiences, enter written and MCQ assessment flows, inspect learning insights, use the tutor, manage profile settings, and request teacher access.

Confirmed from relevant source

Question generation is anchored to the requested subsection while the full section supplies context. Evaluation uses direct section concepts or chapter-level fallback concepts for insight scope when direct links are absent.

Deliberately excluded

Unrelated evaluation, promotion, export, and observability scripts that do not define the learner-facing assessment behavior described here.

Claims about production reliability, latency, provider success rates, or database completeness that would require a separate operational measurement run.

Teacher-dashboard screens not present in the captured evidence set; this report describes the access request and the product's stated institute scope, not an unobserved dashboard implementation.

Recommended acceptance checks

CASE	EXPECTED RESULT
Subsection has direct section concepts	Generate from the subsection with section context; evaluate and attach insights using the direct section concept scope.
Section 10.3 has no direct concepts	Generate from the subsection with section context; evaluate using chapter concepts as fallback; create only evidence supported by the answer.
Unrelated concept supplied	Do not use it to broaden the question or create an unsupported insight. The target subsection and resolved scope remain authoritative.